A-A-53872
24 March 1989

COMMERCIAL ITEM DESCRIPTION

READER, MICROHEMATOCRIT

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-R-37198 which is cancelled.

This commercial item description covers one type of circular microcapillary tube reader, commonly known as a spiral reader, for determining hematocrits directly in terms of percent red cells.

Salient characteristics.

Shall be a spiral reader for use with the glass capillary tube method for hematocrit determinations.

Shall be suitable for determining the percent blood cells (hematocrit) of a centrifuged blood-filled capillary tube.

Shall accommodate capillary tubes with dimensions of 13 to 75 millimeters (mm) in length and 1 to 7 mm in outside diameter.

Shall, as a minimum, consist of a base, two concentric reading disks, and a plastic tube holder. The two reading disks shall be secured to the top center above the base, in a manner that permits free rotation of each disk without binding. The plastic tube holder shall be top mounted, and securely held in position by being fixed between one side of the base and the top center above the base.

Shall have a circular metal base for stability, with at least three equally spaced feet secured to the bottom surface.

Shall have the two concentric reading disks fabricated from corrosion resisting steel. The bottom mounted disk shall have a minimum 8 inch diameter. The top mounted disk shall have a minimum 7 inch diameter.

AMSC N/A FSC 6640

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Shall have bottom mounted disk supplied with an engraved scale around its circumference (for direct reading of percent red blood cells). The scale length shall be a nominal 24 inches with the hematocrit values increasing in a clockwise direction. As a minimum, the scale markings shall include:

Graduation lines for all integers from "2" to "100" percent.
Numbering for each graduation line from "2" to "20" percent.
Numbering for every fifth graduation line above "20" percent.

The disk shall also be fitted with a vertical stop that is positioned to allow the "100" percent graduation line to be aligned with an index line in the plastic tube holder.

Shall have top mounted plate supplied with the following:

An engraved spiral curve on top surface.
Engraved operating instructions on top surface.
A finger hold cutout to facilitate plate rotation.

Shall have tube holder fabricated of a clear and rigid plastic. The spiral curve on the top mounted plate (under the plastic tube holder) shall be clearly visible through the plastic tube holder. The tube holder shall include the following:

A longitudinal groove to hold an inserted capillary tube (i.e. with a blood-filled centrifuged sample).

An index line so located that the bottom of red blood cell column can be properly positioned in the longitudinal groove.

An index line for the initial alignment of the "100" percent graduation line (on the bottom mounted disk) and final hematocrit determination reading (on the bottom mounted disk).

Shall have spiral reader designed to operate in accordance with the following steps:

Step 1. A centrifuged blood-filled capillary tube is inserted in the groove of the plastic tube holder so that bottom of red blood cell column coincides with the appropriate index line in the holder.

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Step 2. The bottom mounted disk is rotated to align the "100" percent graduation line with the appropriate index line in the plastic tube holder.

Step 3. The top mounted disk is independently rotated to align the spiral line with the plasma-air interface within the blood-filled capillary tube.

Step 4. Both disks are rotated together to re-align the spiral line with the red-cell and white cell interface within the blood-filled capillary tube.

Step 5. The percent red cell volume (hematocrit) of the blood sample is then determined by the resultant bottom mounted disk scale reading under the tube holder index line of Step 2.

Shall be constructed so that the base can rest on a flat and level horizontal surface without any evidence of rocking or wobbling.

Workmanship. The microhematocrit reader shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One reader, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

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Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspections and tests. Inspections and tests shall be conducted in accordance with commercial practice to determine compliance with specification requirements.

Sampling for inspections and tests. Sampling for inspections and tests shall be conducted in accordance with commercial practice.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U. S. Food and Drug Administration to be under its jurisdiction, the offeror/ contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

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Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

NOTE: The following National Stock Number is covered by this document:

6640-00-585-1378

Ordering data. (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

PREPARING ACTIVITY:

DoD - MB

Agent:

DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
PHS
FDA-MPQAS

Project No. 6640-1515

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
A-A-53872	Reader, Microhematocrit

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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